

§14.3

The fundamental theorem of line integrals

$C = \{r(t), a \leq t \leq b\}$ a continuous curve, f a continuous and differentiable function. Then

$$\int_C \nabla f(r) \cdot dr = f(r(b)) - f(r(a)) \quad \text{Independent of the path!!!}$$

Proof $\int_C \nabla f(r) dr = \int_a^b [\nabla f(r(t)) \cdot r'(t)] dt =$
 $= \int_a^b \frac{d}{dt} f(r(t)) dt = f(r(t)) \Big|_a^b = f(r(b)) - f(r(a))$

Eg $f(r) = \frac{c}{\|r\|} = \frac{c}{(x^2+y^2+z^2)^{1/2}} \quad \nabla f(r) = -\frac{c r}{\|r\|^3} = -\frac{c}{(x^2+y^2+z^2)^{3/2}} (x, y, z)$

Compute the work done by ∇f along a curve going from $(1, 1, 1)$ to $(4, -1, 2)$

$$W = \int_C \nabla f \cdot dr = f(4, -1, 2) - f(1, 1, 1) = c \left(\frac{1}{\sqrt{21}} - \frac{1}{\sqrt{3}} \right)$$

Path independence theorem

$F(r)$ continuous. Then $\int_C F(r) \cdot dr$ is independent of path if and only if $F(r) = \nabla f(r)$ for some $f: D \rightarrow \mathbb{R}$

If $F(r) = \nabla f(r)$ then $F(r)$ is conservative

Pf $\Leftarrow$ by above thm

$\Rightarrow$ Fix (x_0, y_0) and for any (x, y) , pick C joining (x_0, y_0) & (x, y) . Let $f(x, y) = \int_C F(r) \cdot dr$

Suppose $F = M(x,y) \cdot i + N(x,y) \cdot j$. We must show

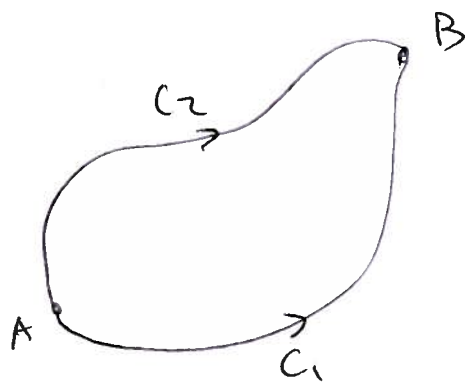
$$\frac{\partial f}{\partial x} = M \quad \& \quad \frac{\partial f}{\partial y} = N$$

$$f(x,y) = \int_{(x_0,y_0)}^{(x,y)} F(r) dr + \int_{(x_0,y)}^{(x,y)} F(r) dr = \int_{y_0}^y N(x_0,t) dt + \int_{x_0}^x M(t,y) dt$$

$$\frac{\partial f}{\partial x} = M(x,y) \quad \square$$

Thm TFAE (1) $F = \nabla f$ (2) $\int F(r) \cdot dr$ indep of path

(3) $\int_C F(r) \cdot dr = 0$ along any closed path



$$\int_{C_1} F(r) \cdot dr = \int_{C_2} F(r) \cdot dr$$

$\Downarrow$

$$\int_{C_1} F(r) \cdot dr - \int_{C_2} F(r) \cdot dr = 0$$

$\Downarrow$

$$\int_{C_1} F(r) \cdot dr + \int_{-C_2} F(r) \cdot dr = 0$$

$\Downarrow$

$$\int_C F(r) \cdot dr = 0$$

$-C_2 =$ curve C_2 with opposite orientation

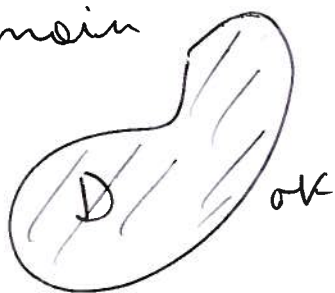
$C =$ Curve given by C_1 followed by $-C_2$.

Theorem Let $F = M_i + N_j + P_k$ M, N, P continuous with continuous derivatives on a ball D .
Then F is conservative iff $\nabla \times F = 0$.

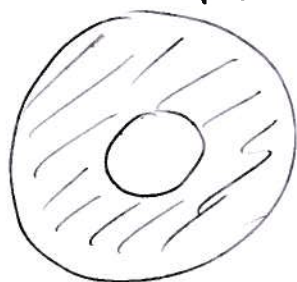
Note If $F = M_i + N_j$ then F is conservative iff $M_y = N_x$

Note The theorem works for any simply connected

domain



not simply connected



Ex $F = (\underset{M}{1+2xy \sin x^2 y}, \underset{N}{1+x^2 \sin x^2 y})$

$$M_y = 2x \sin x^2 y + 2x^3 y \cos x^2 y$$

$$N_x = 2x \sin x^2 y + 2x^3 y \cos x^2 y$$

$$M_y = N_x \text{ so } F = \nabla f; \text{ which } f?$$

~~$$f_x = 1 + 2xy \sin x^2 y + 2x^3 y \cos x^2 y$$~~

$$f_x = 1 + 2xy \sin x^2 y$$

$$f = x + \cos x^2 y + C(y)$$

$$\text{then } f_y = x^2 \sin x^2 y + C'(y) \text{ so } C'(y) = 1 \text{ so } C(y) = y$$

$$f = x + y - \cos x^2 y$$

$$\sum_x F = \underset{M}{(x^3 + 2xy^2, \quad \underset{N}{2x^2y})}$$

$$M_y = 4xy = N_x$$

$$f_y = 2x^2y \Rightarrow f = x^2y^2 + C(x)$$

$$f_x = 2xy^2 + C'(x) \Rightarrow C'(x) = x^3 \Rightarrow C(x) = \frac{x^4}{4}$$

$$f = \frac{x^4}{4} + x^2y^2$$

$$\sum_x F = (-y, x) \quad M_y = -1 \quad N_x = 1 \quad \text{Not conservative}$$

$$\sum_x F = (yz, xz, xy) \quad \nabla \times F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz & xz & xy \end{vmatrix} = \begin{vmatrix} i(x-x) - \\ -j(y-y) \\ +k(z-z) \end{vmatrix} = 0$$

$$f_x = yz \Rightarrow f = xyz + C(y, z)$$

$$f_y = xz + \frac{\partial}{\partial y} C(y, z) \Rightarrow \frac{\partial}{\partial y} C(y, z) = 0 \Rightarrow C(y, z) = C(z)$$

$$f_z = xy + \frac{\partial}{\partial z} C(z) \Rightarrow \frac{\partial}{\partial z} C(z) = 0 \Rightarrow C = \text{const.}$$

$$f = xyz.$$

$\sum_x$ Is F conservative? If so find f such that $F = \nabla f$

$$a) F = (10x - 7y)i + (2y - 7x)j \quad b) F = (12x^2 + 3y^2 + 5y, 6xy - 3y^2 + 5x)$$

$$c) F = \left(\frac{6x^2}{5y^2} \right) i - \left(\frac{4x^3}{5y^3} \right) j \quad d) F = (-e^{-x} \ln y, e^{-x} y^{-1})$$

$\sum_x$ Compute the line integrals. (Note that they are indep of path... why?)

$$a) \int_{(1,2)}^{(3,1)} (y^2 + 2xy) dx + (x^2 + 2xy) dy$$

$$b) \int_{(0,0)}^{(1, \pi/2)} e^x \sin y dx + e^x \cos y dy$$

$$c) \int_{(2,1)}^{(6,3)} \frac{x^3}{(x^4 + y^4)^2} dx + \frac{y^3}{(x^4 + y^4)^2} dy$$